# OOP in Java

Slide 6: Data Types, Comments, and Scanner

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# What are data types

Data types define the type of data a variable can hold.

In Java, there are two categories:

| Primitive Data Types – Basic data types. | Reference Data Types – Objects and arrays. |
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# Primitive Data Types

- **Primitive Data Types in Java:**
  - **byte**: 1 byte (8 bits), range: -128 to 127.
  - **short**: 2 bytes, range: -32,768 to 32,767.
  - **int**: 4 bytes, range: -2^31 to 2^31-1.
  - **long**: 8 bytes, range: -2^63 to 2^63-1.
  - **float**: 4 bytes, used for decimals (e.g., 3.14f).
  - **double**: 8 bytes, for precise decimals (e.g., 3.141592).
  - **char**: 2 bytes, stores a single character (e.g., 'A').
  - **boolean**: 1 bit, stores true or false.

# Explanation

**byte**: Holds small integer values. Example: byte smallNumber = 127;

**short**: A slightly larger range than byte. Example: short mediumNumber = 32000;

**int**: Used for most integer calculations. Example: int largeNumber = 1000000;

**long**: Used for very large integers. Example: long veryLargeNumber = 10000000000L;

**float**: Used for single-precision decimal numbers. Example: float pi = 3.14f;

**double**: More precision for decimal numbers. Example: double precisePi = 3.1415926535;

**char**: Holds a single character. Example: char initial = 'A’;

**boolean**: Can hold true or false. Example: boolean isJavaFun = true;

# Example

```java
// Integer types
byte smallNumber = 127;          // 1 byte, range: -128 to 127
short mediumNumber = 32000;      // 2 bytes, range: -32,768 to 32,767
int largeNumber = 1000000;       // 4 bytes, range: -2^31 to 2^31-1
long veryLargeNumber = 10000000000L; // 8 bytes, use 'L' for long literals

// Floating-point types
float pi = 3.14f;                // 4 bytes, use 'f' for float literals
double precisePi = 3.1415926535; // 8 bytes, more precision than float

// Character type
char initial = 'A';              // 2 bytes, single character, Unicode

// Boolean type
boolean isJavaFun = true;        // 1 bit, can be true or false

// Print all the variables
System.out.println("byte: " + smallNumber);
System.out.println("short: " + mediumNumber);
System.out.println("int: " + largeNumber);
System.out.println("long: " + veryLargeNumber);
System.out.println("float: " + pi);
System.out.println("double: " + precisePi);
System.out.println("char: " + initial);
System.out.println("boolean: " + isJavaFun);
```

# Comments in Java

- What are Comments?
  - Comments help us understand code.
  - They are ignored by the compiler.
- Types of Comments in Java:
  - Single-line comment: Use //
  - Multi-line comment: Use /* ... */
  - Documentation comment: Use /** ... */

```java
// This is a single-line comment
int a = 10; /* This is a multi-line comment
      used to describe the variable */

// Documentation comments are used to explain methods or classes
```

# Scanner for User Input

- **What is Scanner?**
  - The **Scanner** class is used to get input from the user.
  - Import the Scanner class from the java.util package.
    - `import java.util.Scanner;`

UserInputExample.java

```java
import java.util.Scanner;  // Import the Scanner class

public class UserInputExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Getting input from the user
        System.out.print("Enter your age: ");
        int age = scanner.nextInt();  // Read integer input

        System.out.print("Enter your name: ");
        String name = scanner.next();  // Read String input

        System.out.println("Hello, " + name + "! You are " + age + " years old.");
    }
}
```

# Summary

**Data Types**: Different kinds of values variables can hold.

**Comments**: How to write single-line, multi-line, and documentation comments.

**Scanner**: How to take user input and use it in your program.